

4.3.1.11. TXT key for pairing hint (PH)

The optional key **PH** MAY provide a *pairing hint*.

If present, it SHALL be encoded as a variable-length decimal number in ASCII text, omitting any leading zeroes.

The pairing hint represents a base-10 numeric value for a bitmap of methods supported by the Commissionee in its current state for putting it in Commissioning Mode.

For example, the **PH=5** key/value pair represents a hint value with bits 0 and 2 are set.

This value MAY change during the lifecycle of the device.

For example, a device may have a value with bit 0 (Power Cycle) set and bit 2 (Administrator app) unset when in a factory reset state, and then have a value with bit 0 unset and bit 2 set after it has been Commissioned.

The bitmap of methods is defined in [Table 5, “Pairing and Reset Hint Values”](#).

If the Commissionee has enabled [Extended Discovery](#), then it SHALL include the key/value pair for **PH** in the DNS-SD TXT record when not in Commissioning Mode (**CM=0**).

This key/value pair MAY be returned when in Commissioning Mode (**CM=1**).

If the Commissioner does not recognize this value, for example, if the value indicates bit indices defined in a newer version of this specification than the version which the Commissioner implements, then the Commissioner MAY utilize the bits that it does understand and MAY utilize additional data sets available for assisting the user. For example, when a Vendor ID and Product ID are available to the Commissioner, the [Section 11.23, “Distributed Compliance Ledger”](#) may also provide a URL for the Device User Guide which can contain additional information to help in Commissioning this Commissionee.

Some of the pairing hints SHALL or MAY require additional information to be encoded for proper expression of their meaning. This is accomplished with the **PI** TXT key, described in a following section. Dependency on usage of the **PI** key is expressed by the **Instruction Dependency** column in the table below. For some entries, the use of the **PI** key is mandatory (indicated as **M**), while for some other entries, the use of the **PI** key is optional (indicated as **O**). For the remaining entries the use of the **PI** key does not apply (indicated as **no**).

The fields in the bitmap are defined for values of the **PH** key in the [Pairing and Reset Hint Values](#) table below. This bitmap describes methods for putting the device into a specific target state. It is referenced here to describe methods for putting the device into Commissioning Mode, and it is referenced elsewhere to describe methods for putting the device into other states, such as the Factory Reset state in the DCL DeviceModel Schema’s [FactoryResetStepsHint](#) field.

Table 5. Pairing and Reset Hint Values